DEPARTMENT OF THE AIR FORCE
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FULL-SIZED VAN AND SPORT UTILITY VEHICLE

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QUALIFICATION TRAINING PACKAGE

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Chapter 1

OVERVIEW

1.1. Overview. This Qualification Training Package (QTP) establishes standardized training requirements, content, and certification protocols for all Air Force and Space Force operators of this vehicle. Designed to ensure consistency and excellence across all installations, this package provides comprehensive guidance for trainers, trainees, and certifiers throughout the certification process.

1.1.1. Instructional Standards. The training standards outlined herein align with Department of the Air Force (DAF) requirements and serve as the authoritative source for curriculum development and performance evaluation. Trainers will utilize **Chapters 2** through **Chapter 6** to guide trainee preparation, conduct classroom lectures, provide demonstrations, and administer knowledge and performance evaluations.

1.2. Administrative. Send comments and suggested improvements on DAF Form 847, *Recommendation for Change of Publication*, through Air Force Installation and Mission Support Center functional managers via e-mail at: AFIMSC.A34OG.GroundTrans@us.af.mil.

Chapter 2

RESPONSIBILITIES AND PREPARATION

2.1. Trainer. The trainer is responsible for the overall execution of the training plan. This includes providing an overview of **Chapters 2** and **3**, delivering classroom lectures based on **Chapter 4**, and conducting knowledge training and demonstrations (**Chapter 5**). The trainer will administer and grade all review questions, ensuring that any missed items are thoroughly reviewed with the trainee to guarantee task comprehension before signing off on the training block.

2.2. Trainee. The trainee must actively participate in the QTP process and meet all preparation requirements before beginning practical instruction.

2.2.1. Administrative Requirements: The trainee must possess a valid state driver's license and ensure Sections I and II of the DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License,* are completed in accordance with (IAW) Department of the Air Force Instruction 24-301, *Ground Transportation*.

2.2.2. Mandatory Reading: The trainee must read this entire QTP and the manufacturer's operator's manual for the specific vehicle being trained on.

2.2.3. Execution: During instruction, the trainee will follow along with the lecture, utilize **Attachment 2** and **Attachment 4** as inspection guides, and prepare for the unassisted performance test.

2.3. Certifier. The certifier acts as the independent evaluating authority. They are responsible for evaluating the Airman's physical task performance (**Chapter 6**) without providing any assistance or coaching. Upon successful, independent completion of the performance test, the certifier will sign off on the task(s) to finalize the licensing process.

Chapter 3

INTRODUCTION

3.1. Objectives. The objective of this course is to ensure personnel become fully qualified van and/or large sport utility vehicle (SUV) operators who possess the requisite knowledge and skills for safe, professional execution IAW Department of Defense Instruction 6055.04, *DoD Motor Vehicle and Traffic Safety*. Through a comprehensive curriculum of lectures, demonstrations, and hands-on driving sessions, trainees will be trained in both the safe operation and preventive maintenance of various oversize cargo truck platforms. Ultimately, trainees must demonstrate their proficiency by independently performing operator inspections and completing the final performance test with zero instructor assists.

3.2. Desired Learning Outcomes.

3.2.1. Understand the safety precautions to be followed before-, during-, and after-operation of the full-size van.

3.2.2. Understand the purpose of the full-size van and its role in the mission.

3.2.3. Know the proper operator maintenance procedures of the full-size van IAW applicable technical orders (TO) and use of Department of Air Force (DAF) Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.2.4. Safely and proficiently operate the full-size van.

3.3. Lesson Duration.

3.3.1. Recommended instructional and hands on training time is 3.25 hours:

Table 3.1. Recommended Training Time for Training Activities. **Note:** This is a recommended time; training time may be more or less depending how quickly a trainee learns new tasks.

Training Activity	Training Time
Trainee's Preparation	30 Minutes
Instructor's Lecture and Demonstration	45 Minutes
Trainee's Personal Experience (to build confidence and proficiency) ▪ Perform Operator Maintenance ▪ Operate the Vehicle	1 Hour
Trainee's Performance Evaluation	1 Hour

3.4. Instructional References.

3.4.1. Risk Management (RM) and Safety Principles.

3.4.2. Applicable TOs or Manufacturer's Operator's Manual (see vehicle maintenance for TO number for vehicle being used in training).

3.4.3. DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.5. Instructional Training Aids and Equipment.

3.5.1. Full-Size van Operator Training Package.

3.5.2. Full-Size van and large SUVs.

3.5.3. Applicable TO or manufacturer's operator's manual.

3.5.4. DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

3.5.5. Videos (if locally produced).

3.5.6. Suitable training area.

3.5.7. Traffic cones or suitable markers.

Chapter 4

KNOWLEDGE OVERVIEW AND OPERATIONAL REQUIREMENTS

4.1. Knowledge Overview. Operating a full-size van or a large SUV demands a higher level of skill, spatial awareness, and safety consciousness than driving a standard passenger sedan. These vehicles are characterized by their large wheelbases, elevated center of gravity, and high weight capacity.

4.1.1.The DAF utilizes a diverse fleet of full-size vans and large SUVs to carry out critical transport missions. Because these vehicles are primarily intended to carry numerous passengers alongside cargo, your role as an operator carries an immense responsibility: ensuring the safety of your team and the integrity of your equipment.

4.2. Overview of Training and Requirements.

4.2.1. Training Objectives. This training package is designed to qualify you to safely operate and maintain the DAF's van and SUV fleet. At the end of this instructional block, you will be expected to:

4.2.1.1. Inspect and Verify: Perform thorough operator inspections and complete a physical performance test with zero instructor assists, utilizing lectures, demonstrations, and hands-on driving practice.

4.2.1.2. Operate Safely: Master defensive driving techniques and preventive maintenance habits tailored specifically to larger vehicles.

4.2.1.3. Represent as a Professional: Develop the baseline skills required of a qualified, professional operator who consistently executes missions safely and efficiently.

4.2.2. Desired Learning Outcomes. To successfully qualify, trainees must demonstrate a firm understanding of:

4.2.2.1. Lifecycle Safety: The exact safety protocols required before, during, and after vehicle operation.

4.2.2.2. Mission Integration: The specific role these transport vehicles play within your Unit, Base, the local community (such as during natural disaster response), and the broader Air Force mission.

4.2.2.3. Regulatory Compliance: Proper operator maintenance in accordance with applicable Technical Orders (TOs) and the correct usage of DAF Form 1800, *Operator's Inspection Guide and Trouble Report.*

4.2.2.4. Operational Proficiency: Seamless vehicle control that meets mission timelines while preserving passenger safety.

4.2.3. Vehicle Profile & Specifications. Full-size vans and SUVs vary widely in dimensions

and capabilities depending on their make, model, and year. Before turning the key, you must know your specific vehicle's limits. While this information is detailed in the vehicle's Technical Order (TO), you can quickly verify vital limits using the Vehicle Data Plate located inside the driver-side door jamb.

4.2.4. Operating the Four-Wheel Drive (4WD) System. Many military-spec large SUVs and vans feature 4WD systems. When engaged, power is distributed to all four wheels through a central transfer case or power transfer unit. This drastically increases traction, allowing you to navigate snow, mud, loose gravel, or off-road conditions that would stall a standard rear-wheel-drive vehicle.

4.2.5. When shifting from two-wheel drive (2WD) to 4WD while in motion, you may hear a momentary "clunk" or ratcheting sound. This is normal; it is simply the front drivetrain components accelerating to match the speed of the rear axle. Do not let this alarm you. However, always consult the manufacturer's operator manual for vehicle-specific shifting limits and maximum speeds in 4WD modes.

4.3. Vehicle Inspection. Routine inspections prevent minor issues from turning into major breakdowns or accidents. If you find any discrepancies during any phase of your inspection, immediately report them to your Vehicle Control Official, your supervisor, and/or vehicle maintenance.

4.3.1. Phase 1: Pre-Trip Inspection. Before starting the vehicle, walk around it using the systematic Seven-Step Inspection Method (found in detail in Attachment 4) alongside DAF Form 1800, *Operator's Inspection Guide and Trouble Report* and **Attachment 2**. Your walk-around must verify:

4.3.1.1. Fluids & Leaks: Look underneath the chassis for active leaks (fuel, oil, coolant, hydraulic fluid, or air). Open the hood and verify that engine oil, coolant, power steering, transmission, and windshield washer fluids are within normal operating limits.

4.3.1.2. Belts & Hoses: Inspect drive belts for proper tension, fraying, or cracking. Ensure coolant and heater hoses are securely connected and free of dry rot.

4.3.1.3. Battery: Check that the battery is mounted securely, connections are tight, and terminals are free of corrosion.

4.3.1.4. Tires & Wheels: Inspect all tires for proper inflation, sidewall damage, bulges, cuts, and adequate tread depth. Ensure wheel rims are free of cracks or splits and verify that all lug nuts are present and tight.

4.3.1.5. Exterior & Visibility: Inspect the windshield, windows, and mirrors for cracks or dirt. Test the windshield wipers and washer fluid spray. Ensure the fuel door and cap are intact and secure.

4.3.1.6. Electrical & Signals: Turn on all lights—including headlights, high beams, taillights, brake lights, turn signals, emergency flashers, and clearance/marker lights—to verify operation.

4.3.1.7. Cabin Safety: Test the horn, heater, defroster, and control panel gauges. Ensure all doors latch securely. Adjust all mirrors to eliminate blind spots and inspect every seatbelt for fraying and proper buckle engagement.

4.3.2. Phase 2: During-Operation Monitoring. Your inspection doesn't stop once you put the vehicle in gear. While driving, maintain constant situational awareness of your vehicle's mechanical health:

4.3.2.1. Scan the Instrument Panel: Routinely check your warning lights and instrument gauges, paying close attention to oil pressure, coolant temperature, system voltage, and fuel level.

4.3.2.2. Listen for exhaust leaks, unusual engine knocks, transmission whining, or air leaks.

4.3.2.3. Smell for burning rubber, hot oil, electrical shorts, or fuel fumes.

4.3.2.4. Feel for abnormal steering wheel vibrations, pulling to one side, or spongy brake pedal response.

4.3.3. Phase 3: After-Operation Procedures. Once the mission is complete, perform these wrap-up steps to ensure the vehicle is ready for the next crew:

4.3.3.1. Cleanliness: Remove all trash, dirt, and personal gear. Ensure the interior and exterior are clean.

4.3.3.2. Refuel/Recharge: Top off the fuel tank or return the vehicle to a designated charging station.

4.3.3.2.1. Safety First: Always turn off the engine, set the parking brake, and do not smoke or use electronic devices while refueling. Never refuel inside a closed building.

4.3.3.3. Secure the Vehicle: Park in a designated spot, set the parking brake, roll up the windows, and lock the doors. Record any final discrepancies and your ending mileage on DAF Form 1800, *Operator's Inspection Guide and Trouble Report.*

4.4. Safety Equipment and Human Factors. Operating a large passenger vehicle introduces distinct hazards. Understanding these risks allows you to mitigate them before they turn into mishaps.

4.4.1. The Four Common Mishap Types

4.4.1.1. Blind Spots: Large vans have massive blind spots along the sides and directly behind the vehicle. Standard rearview mirrors do not cover these zones completely.

4.4.1.2. Backing Accidents: Because rear visibility is restricted, backing is the single most common source of fleet vehicle damage.

4.4.1.3. Overhead Clearance: Low-hanging tree branches, parking garages, and drive-thru

canopies present severe collision risks for vehicles standing over 7 feet tall.

4.4.1.4. Rollovers: A high center of gravity combined with passenger weight drastically increases rollover risk during sudden evasive maneuvers or high-speed cornering.

4.4.2. Required Safety Gear. Always ensure your vehicle is stocked with the necessary safety equipment before departing:

4.4.2.1. High-Visibility Gear: Reflective vests or belts must be worn during low-visibility operations and at all times when operating on the flightline.

4.4.2.2. Inclement Weather Gear: Prepare for sudden weather changes by carrying appropriate cold-weather, rain, or heat-mitigation gear.

4.4.2.3. Emergency Kit: Verify the presence of a functional first aid kit, emergency warning triangles, and any hearing protection required for high-noise airfield zones.

4.5. Driving Safety and Tactical Precautions.

4.5.1. General Operating Safety Rules

4.5.1.1. Never Exceed Passenger Limits: Do not allow more passengers than there are functional seatbelts. Every rider must be buckled up prior to movement. No riders are allowed to ride anywhere other than the designated passenger seats (never in cargo areas or truck beds).

4.5.1.2. Load Balance: Distribute passenger and cargo weight evenly. Avoid loading cargo onto the roof, as this raises the vehicle's center of gravity and exacerbates rollover risk.

4.5.1.3. Control Your Speed: Adjust your speed downward for heavy loads, dense traffic, or inclement weather.

4.5.1.4. Blown-out Tire Protocol: If a tire blows out, remain calm. Hold the steering wheel firmly with both hands, ease off the accelerator, and let the vehicle naturally slow down. Do not slam on the brakes. Safely steer onto the road shoulder.

4.5.1.5. Unattended Vehicles: Never leave a vehicle running unattended. Turn off the engine, remove the keys, and set the parking brake whenever you exit the driver's seat.

4.5.1.6. Share the Road: Always yield the right of way to emergency vehicles (ambulances, fire engines, security forces) utilizing lights and sirens.

4.5.2. Flightline-Specific Foreign Object Damage (FOD) Protocols. FOD is a critical hazard to aircraft engines. As a vehicle operator, you are the first line of defense against flightline debris.

4.5.2.1. Entry Inspections: Before entering any airfield environment, you must conduct a physical inspection of your tires and treads, picking out any rocks, screws, or debris. The

only exception to this rule is emergency response vehicles actively responding to a real-world emergency.

4.5.2.2. Off-Road Recovery: If your vehicle is driven on an unpaved surface, you must perform a thorough tire FOD check immediately before re-entering the paved airfield area.

4.5.2.3. Required Tools: Fleet vehicles that frequently access the flightline must be equipped with:

4.5.2.3.1. A FOD Picker (which must be etched with the vehicle's specific identification number).

4.5.2.3.2. A secure, covered FOD Container labeled clearly with the letters "FOD" and emptied daily.

4.5.3. Emergency Tire Changing Safety. Changing a flat tire on a heavy van or SUV can be incredibly hazardous. Follow this tactical safety checklist to protect yourself:

4.5.3.1. Evaluate the Location: If you get a flat tire on a bridge, sharp curve, or a busy road with no shoulder, do not stop. It is far safer to ruin a tire by slowly driving to a safe, wide-open location than to risk being struck by traffic.

4.5.3.2. Prep the Vehicle: Park on a firm, level surface. Turn on your four-way hazard flashers, place the vehicle in "Park" (or low gear for manual transmissions), set the parking brake, and shut off the engine.

4.5.3.3. Secure the Area: Place emergency warning triangles or flares behind your vehicle to alert oncoming drivers.

4.5.3.4. Chock the Wheels: Block the tire diagonally opposite the flat. If changing a front tire, block the rear wheels. If changing a rear tire, block the front wheels.

4.5.3.5. Jack Setup: Ensure your jack is rated for the vehicle's gross weight. Position the jack on a solid, load-bearing portion of the frame (refer to the owner's manual).

4.5.3.6. Safety First:

4.5.3.6.1. If available, always slide jack stands under the frame once lifted.

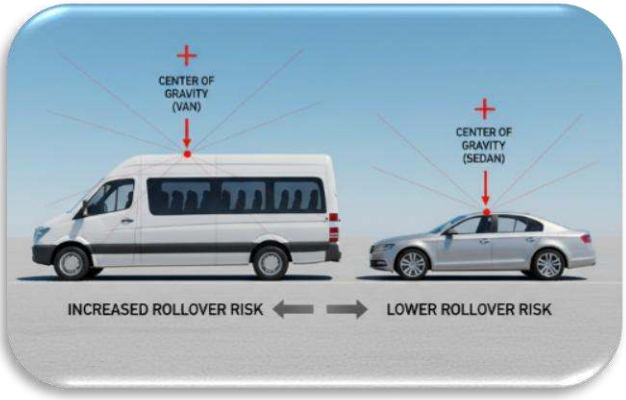
4.5.3.6.2. Never put any part of your body under a vehicle supported only by a jack.

4.5.3.6.3. Double-check that all lug wedges are loose before completely removing lug nuts.

4.5.3.7. Stow and Secure: Once the tire is changed, secure the damaged tire and jack tools in their designated onboard compartment. Unsecured jacks and spare tires can become lethal flying projectiles in a crash. Return the repaired tire to its primary storage mount as soon as possible.

4.6. High Rollover Risk. The physical profile of full-size vans and large SUVs places them at high risk for rollover crashes.

Figure 4.1. Center of Gravity Depiction.



4.6.1. Rollover Physics & Statistics

4.6.1.1. The Danger: Vehicles with a high gross weight and high center of gravity (such as vans, buses, wreckers, and large tractors) require much longer to stabilize during lateral shifts.

4.6.1.2. The Reality: According to Department of Transportation data, rollover crashes are incredibly violent and represent a leading cause of highway fatalities. Unrestrained occupants in a rolling vehicle are subjected to catastrophic forces. In 2022, nearly 9,000 passenger vehicle occupants were killed in rollover crashes—accounting for approximately 40% of all light vehicle occupant deaths.

4.6.1.3. Preventability: These crashes are preventable. Rollovers are almost always the result of driving too fast for road conditions, poor tire maintenance, incorrect cargo loading, or abrupt steering corrections ("over-correcting").

4.6.2. Rollover Prevention Techniques. To keep your vehicle firmly on its tires, you must practice these habits:

4.6.2.1. Smooth Steering: Avoid abrupt lane changes or sharp, rapid steering maneuvers. Keep your lane transitions gradual, checking that you have ample clearance in front and behind.

4.6.2.2. Control Speed on Curves: Always slow down before entering a curve or highway off-ramp. Never accelerate into a sharp turn.

4.6.2.3. Strict Seatbelt Discipline: Enforce a strict "one person per seatbelt" policy. Seatbelts are your primary defense against ejection and death during a rollover event.

4.6.2.4. Never Overload: Strictly respect passenger limits and distribute heavy gear along the floor of the vehicle rather than high up.

4.6.2.5. Maintain Tires: Keep your tires properly inflated and check them daily. Under-inflated tires can flex under lateral load, causing the tire bead to break and triggering a rollover.

4.7. Vehicle Operation and Maneuvering

4.7.1. Steering Control

4.7.1.1. Hand Position: Always maintain two hands on the steering wheel in a balanced position (9 and 3 o'clock or 10 and 2 o'clock). This gives you the leverage needed to recover from sudden road hazards or tire blowouts.

4.7.1.2. Wider Turn Arcs: Because of the extended wheelbase of a van or large SUV, you must swing slightly wider into turns than you would in a sedan to prevent the rear wheels from striking curbs, signs, or pedestrians.

4.7.2. Executing Safe Turns.

4.7.2.1. The Mirror Sweep: Before executing any turn, systematically check your traffic to the front, sides, and rear. Check your side mirrors first, paying close attention to the side closest to the direction of your turn.

4.7.2.2. Over-the-Shoulder Blind Spot Checks: Mirror glass cannot show you everything. Because you sit higher in a van, passenger cars and pedestrians can easily hide in your blind spots. Turn your head and perform a physical blind-spot check before changing lanes or turning.

4.7.3. Braking Dynamics. Stopping a vehicle weighing nearly 10,000 lbs. takes significantly more time and distance than stopping a standard sedan. Your stopping distance is dictated by several critical factors:

4.7.3.1. Speed & Weight: Doubling your speed quadruples the kinetic energy your brakes must convert to heat. A fully loaded 15-passenger van requires nearly twice the stopping distance of an empty van.

4.7.3.2. Environmental Conditions: Rain, snow, ice, or loose gravel dramatically reduce tire traction. Double your following distance in poor weather.

4.7.3.3. Mechanical Readiness: Worn brake pads or thin tire tread can cause your brakes to fade or lock up, increasing stopping distances exponentially.

4.7.4. Precision Backing and Spotter Protocols. Whenever possible, plan your route and parking to eliminate or minimize the need to back up. If backing is unavoidable, you must adhere to the following protocol:

4.7.4.1. Use a Spotter: If a spotter is available, you must use one. The spotter must position themselves outside and to the rear of the vehicle, where they can see both the path of the vehicle and the driver's face in the side mirror.

4.7.4.2. Maintain Constant Visuals: You must always keep visual contact with your spotter in your mirrors. If you lose sight of your spotter for even a split second, you must stop the vehicle immediately.

4.7.4.3. Sound the Horn: Give a brief tap of the horn before beginning to back up to ensure your spotter and bystanders know the vehicle is moving.

4.7.4.4. Go Slow: Back at a crawl, utilizing both side mirrors to monitor clearance.

4.7.5. Passenger Transport Protocols. When your mission involves carrying personnel, your safety habits must be flawless:

4.7.5.1. Pre-boarding Safety Check: Ensure the cabin is clean, seatbelts are fully accessible and functional, and cargo is safely lashed down.

4.7.5.2. Curbside Loading/Unloading: Always load and unload your passengers on the curb side of the vehicle whenever possible to keep them out of active traffic lanes.

4.7.5.3. Activate Hazards: Use your hazard lights during passenger boarding and exiting to warn surrounding traffic.

Chapter 5

EXPLANATION AND DEMONSTRATION

5.1. Trainer Preparation and Safety Protocols. Prior to instruction, the trainer must establish a safe training location, reserve an appropriate vehicle, obtain the manufacturer's operator's manual, and ensure the trainee has completed a DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License*. Both trainers and trainee must adhere strictly to safety protocols, including the removal of jewelry and identification tags, the use of wheel chocks when parked, and the mandatory use of seatbelts. The trainer should conduct an initial walk-around to familiarize themselves and the trainee with all warning labels, data plates (specifically noting load capacities and rollover risks), and vehicle dimensions.

5.1.1. Personal Protective Equipment and RM. Trainers and trainees must wear appropriate PPE, including safety steel-toed boots, gloves for cargo handling, and a reflective belt during hours of darkness or on the flightline. Throughout the demonstration, the trainer will actively model the five-step Air Force RM process, specifically assessing hazards related to heavy cargo loads, high centers of gravity, and overhead clearances.

5.2. Operator Maintenance and Vehicle Pre-Trip Inspection. The trainer will accomplish a comprehensive vehicle inspection alongside the trainee, documenting it on the DAF Form 1800, *Operator's Inspection Guide and Trouble Report*. The trainer will demonstrate the mandated seven-step inspection method (**Attachment 4**) and utilize the inspection guide (**Attachment 2**). Crucially, the trainer will demonstrate how to conduct a proper tire Foreign Object Damage (FOD) check and, if the truck is equipped with air brakes, perform the comprehensive final in-cab air brake tests (testing the air leakage rate, the low-air pressure warning signal, and the emergency spring brake pop-out).

5.3. Operational Demonstration. Utilizing the vehicle data plate and the manufacturer's operator's manual, the trainer will explain the vehicle's capacities, controls, standard air pressure operating ranges, and all warning lights/buzzers. The trainer will demonstrate steering, turning, changing lanes, and navigating varying grades.

5.3.1. Pulling Away and Slow-speed Braking: Confirming the brakes bite immediately.

5.3.2. Steering & Wide-Arc Turning: Showing how to execute turns by swinging slightly wider than a standard sedan, preventing the rear tires from jumping curbs or clipping obstacles due to the vehicle's long wheelbase.

5.3.3. Blind Spot Mirror Sweeps: Demonstrating the "mirror sweep" prior to changing lanes, emphasizing how easily small passenger cars disappear in the blind spots of a 15-passenger van or large SUV.

5.3.4. Overhead Clearance: The trainer will drive the vehicle near a simulated clearance hazard, showing how to cross-reference the vehicle's height limits (e.g., 7'0") with overhead structural markings.

5.3.5. Internal Passenger Load Balance: The trainer will explain the critical risk of "rear-end

loading." They will demonstrate how to pack cargo along the floorboards and keep passengers distributed evenly across the front and center rows before loading the final rear bench, preserving the vehicle's structural center of gravity.

5.3.6. Specialized Systems: If equipped with 4WD, the trainer will demonstrate the proper shifting procedures for the transfer case (H2, H4, N, and L4) while in motion and stationery. The trainer will demonstrate the exact speed ranges permitted during shift-on-the-fly execution and show how the vehicle must be completely stopped with the transmission placed in Neutral before shifting into 4L (Low Range).

5.3.7. Backing Procedures. Due to the size and blind spots, DAF regulations always require the use of a spotter when backing. The trainer will review standard hand signals and have the trainee serve as the spotter, demonstrating how to always keep the spotter in sight in the mirrors, and stopping the vehicle immediately if visual contact is lost.

5.4. Post-Operation Demonstration. Following the driving portion, the trainer will demonstrate after-operation procedures. This includes cleaning the vehicle, properly stowing cargo straps and chains, draining the air tanks (mandatory for air brake equipped vehicles to prevent moisture buildup), refueling, and executing the manufacturer's shut-down procedures. The trainer will park the vehicle in a level area, place the transmission in neutral (or park), apply the parking brake, and perform a post-operation walk-around, annotating any discrepancies on the DAF Form 1800, *Operator's Inspection Guide and Trouble Report*.

Chapter 6

TRAINEE PERFORMANCE AND EVALUATION

6.1. Trainee Performance and Practical Application. Following the demonstration, the trainee will transition to behind-the-wheel practice. The trainer must ensure safety at all times and immediately stop training if any safety infractions occur. If available, the vehicle should be marked with a "Driver in Training" sign. The trainee must don appropriate PPE (e.g., safety steel-toed boots, gloves for cargo handling, and reflective belts during low visibility or on the flightline), fasten their seatbelt, and properly adjust the driver's seat and all mirrors to account for the vehicle's substantial blind spots before moving.

6.2. Trainee Vehicle Inspection Execution. The trainee will conduct operator maintenance and pre/during inspections, utilizing **Attachment 2** as a guide. The trainee must verbally explain the items being inspected to the trainer, including the execution of a tire Foreign Object Damage (FOD) check. The trainee must accurately verify all fluid levels, belts, hoses, tire condition, safety equipment, and warning lights without trainer assistance, ensuring the DAF Form 1800, *Operator's Inspection Guide and Trouble Report* is properly documented.

6.3. Trainee Operational Execution. The trainer will establish a road course containing turns, intersections, and varying terrain. The trainee will practice operating the full-size van or large SUV forward and backward until they show proficiency. The trainee must successfully explain the vehicle's passenger/cargo load capacities, overhead clearance restrictions, instrument panel warnings, and rollover mitigation techniques.

6.3.1. Operations. They will practice wide-arc turns, over-the-shoulder blind spot checks, lane changes, entering/exiting highways, and speed control. The trainee must demonstrate smooth steering and gradual braking to accommodate the vehicle's heavy weight and high center of gravity, effectively mitigating rollover risks.

6.3.2. Passengers, Cargo & Systems. The trainee must demonstrate the ability to properly load and balance passengers and cargo along the floorboards to maintain a stable center of gravity and prevent rear-end loading. They must also execute safe curbside passenger loading/unloading procedures using hazard lights. If the vehicle is equipped with a 4-Wheel Drive (4WD) system, the trainee must successfully demonstrate safely shifting through all 4WD ranges (e.g., H2, H4, N, and L4) in accordance with the manufacturer's speed and operational limits.

6.3.3. Backing Practice. The trainee will practice backing maneuvers utilizing the trainer (or another available trainee) as a spotter. The trainee must strictly adhere to sounding the horn before moving, using standard hand signals, maintaining constant visual contact with the spotter via the mirrors, and stopping the vehicle immediately if the spotter disappears from view.

6.4. Performance Evaluation. Once proficient, the trainee will complete the official performance evaluation using the checklist found in Attachment 3. The trainer will review Attachment 3 with the trainee and answer any final questions. During the evaluation, the trainer will ensure all safety, PPE, passenger safety protocols, load balancing, and operator manual warnings are strictly

observed. The trainee must complete the evaluation checklist with zero instructor assistance.

6.4.1. Retraining and Re-evaluation. If the trainee fails any portion of the evaluation, they will be issued a "No-Go." The trainer will critique the observed weaknesses and re-demonstrate the failed items. The trainee must re-perform the specific tasks until they show proficiency before an official re-evaluation is conducted.

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DoDI 6055.04, *DoD Motor Vehicle and Traffic Safety*, 27 August 2021

DAFI 24-301, *Ground Transportation*, 9 January 2026

Adopted Forms

DAF Form 171, *Request for Driver's Training and Addition to U.S. Government Driver's License*

DAF Form 847, *Recommendation for Change of Publication*

DAF Form 1800, *Operator's Inspection Guide and Trouble Report*

Abbreviations and Acronyms

4WD—Four-Wheel Drive

ABS—Anti-lock Braking System

DAF—Department of the Air Force

FOD—Foreign Object Damage

IAW—In Accordance With

PPE—Personal Protective Equipment

QTP—Qualification Training Package

RM—Risk Management

SUV—Sport Utility Vehicle

TO—Technical Order

Terms

Center of Gravity—The point in a vehicle where its weight is evenly distributed. A higher center of gravity, which is characteristic of vans and large SUVs, reduces vehicle stability and significantly increases the risk of a rollover, especially during turns or sudden maneuvers.

Foreign Object Damage—Damage caused to aircraft engines or equipment by objects such as rocks, screws, or other debris. Vehicle operators on an airfield are responsible for conducting regular tire inspections to prevent FOD.

Rollover—A type of vehicle crash where a vehicle tips over onto its side or roof. Due to their high center of gravity, full-size vans and large SUVs are at a higher risk for rollovers, which are often caused by excessive speed in turns, abrupt steering, or improper loading.

Spotter—A person who is required to guide a driver during backing maneuvers. The spotter positions themselves outside the vehicle to see the vehicle's path and maintain constant visual contact with the driver in the mirrors to ensure safe movement.

Attachment 2

VEHICLE INSPECTION GUIDE

Figure A2.1. Vehicle Inspection Guide.

Inspection Guide - Oversized Cargo Truck

STEP 1. Vehicle Overview

- ☐ **Paperwork**
  - DAF Form 1800
- ☐ **Vehicle Approach**
  - Damage
  - Vehicle Leaning
  - Fluid Leaks (puddling on ground/frame)
  - Surrounding Hazards

STEP 2. Engine Compartment

- ☐ **Leaks/hoses/Electrical Wiring Insulation**
- ☐ **Oil Level**
- ☐ **Coolant Level**
- ☐ **Power Steering Fluid**
- ☐ **Windshield Washer Fluid**
- ☐ **Battery Connections & Tie-downs**
- ☐ **Automatic Transmission Fluid Level**
- ☐ **Engine Compartment Belts**

STEP 3. Engine Start/ Cab Check (Left/Front/Right)

- ☐ **Safe Start**
- ☐ **Gauges**
  - Oil Pressure Gauge
  - Air Pressure Gauge
  - Temperature Gauge (Coolant/Engine Oil)
  - Ammeter/Voltmeter
- ☐ **Warning Lights & Buzzers**
- ☐ **Mirrors & Windshield**
- ☐ **Wipers/Washers**
- ☐ **Emergency & Safety Equipment**
  - Red Reflective Triangles
  - Properly Charged & Rated Fire Extinguisher
  - Optional (Chains/Tire Changing Equip, Emergency Phone List)
- ☐ **Lights/Reflectors/Reflector Tape**
  - Dash Indicators for:
    - Left Turn Signal
    - Right Turn Signal
    - Four-Way Emergency Flashers
    - High Beam Headlight
    - ABS Indicator (If equipped)
    - Clearance Lights
  - Reflective Clean/Functional Light/Reflector Checks Include:
    - Headlights (clear)
    - Taillights (red)
    - Backing Lights (clear)
    - Turn Signals (amber, except in rear)
    - Four-Way Flashers (amber, except in rear)
    - Brake Lights (red)
    - Red Reflectors & Amber Reflectors (side/rear)
    - Reflective Tape Condition
- ☐ **Horn**
- ☐ **Heater/Defroster**
- ☐ **Brakes**
  - Parking Brake Check
  - Hydraulic Brake Check
  - Air Brake Check (if equipped)
  - Service Brake Check
- ☐ **Safety Belt**



STEP 4. Walk-Around Inspection

- ☐ **Steering**
  - Steering Box/Hoses/linkages
- ☐ **Suspension**
  - Springs/Air/Torque/Mounts
  - Shock Absorbers
- ☐ **Brakes**
  - Slack Adjustor's & Push-rods
  - Brake Chambers
  - Brake Hoses/Lines
  - Drum Brake/Brake Linings
- ☐ **Wheels**
  - Rims
  - Tires
  - Hub Oil Seals/Axle Seals
  - Lug Nuts
  - Doors
  - Mirrors
  - Fuel Tank

Attachment 3

PERFORMANCE TEST

A3.1. Instructions. Before beginning the performance test, the trainer will brief the trainee on the scenario he/she will need to accomplish. The trainees will be given additional directions and instructions as needed as they proceed through the scenario.

A3.2. Scoring.

A3.2.1. The trainer examiner will be scoring the trainee on full-size van operations and also the general safe driving practices. The examiner will give directions and instructions to the trainee in sufficient time to execute a driving maneuver. The trainee will not be asked to drive in an unsafe manner.

A3.2.2. The examiner will be making various marks on the performance test checklist. This does not necessarily mean they have done anything wrong. It is in the best interest to concentrate on the operation of the full-size van. The trainer will explain the test results to the trainee at the conclusion of the performance test.

A3.2.3. Tasks being graded are listed on the following page; the trainee will be required to successfully pass all items.

A3.2.4. The instructor will stop the test at any time safe full-size van operations are not being followed or as deemed necessary for safety concerns.

Figure A3.1. Performance Test Checklist.

PERFORMANCE TEST CHECKLIST

Instructions to Certifier and Trainee:

Pre-Test Briefing: The Certifier will review this entire checklist with the trainee and brief the intended road course route. The trainee must understand that all driving directions will be provided in sufficient time to execute maneuvers safely.

Scoring Awareness: The Certifier will make various marks on this checklist during the evaluation. These marks do not inherently indicate a "No-Go"; the trainee must focus exclusively on safe vehicle operation.

Conduct of Test: No Trainer or Certifier assists are permitted during the evaluation. Results will be explained to the trainee upon the conclusion of the test.

Safety: The Certifier will stop the test immediately if unsafe operations occur or health concerns arise.

SECTION I. TRAINEE AND CERTIFIER INFORMATION

1.Trainee Name (Last, First MI)	**2. Unit/Office Symbol**
3. Certifier Name (Last, First MI)	**4. Unit/Office Symbol**

SECTION II. PERFORMANCE TEST INFORMATION

5. Vehicle Management Code for Vehicle Tested	**6. Date**

SECTION III. SCORESHEET

7. Equipment & Posture	Go	No Go
Verifies vehicle required forms & personal protective equipment	☐	☐

8. Pre-Operational & General Safety:	Go	No Go
Conducts 360 degree safety inspection when approaching veh	☐	☐
Inspects items identified on DAF1800, documents discrepancies	☐	☐
Conducts ops check of controls & instruments	☐	☐
Sign DAF1800	☐	☐
Performs post trip inspection,reports any discrepancies to Veh Mgt	☐	☐

9. On-Road Operational Compliance:	Go	No Go
Obeys all traffic signs, signals, local/military traffic laws.	☐	☐
Completes course without accidents/moving violations.	☐	☐
Regular traffic/mirror checks/safe following distances.	☐	☐
Uses proper signals/smooth, safe lane changes.	☐	☐

10. Maneuvers (Turns and Curves):	Go	No Go
Turns: Checks traffic in all directions/signals properly.	☐	☐
Turns: Decelerates smoothly/proper gear for the turn.	☐	☐
Turns: Checks mirrors/avoids drifting into oncoming traffic.	☐	☐
Curves: Reduces speed/proper gear prior to entry	☐	☐
Curves: Maintains lane positioning/360-degree traffic checks	☐	☐

11. Intersections and Hazards:	Go	No Go
Int.: Decelerates gently/brakes smoothly on approach.	☐	☐
Int: Stop (no coasting) behind stop lines/signs/sidewalks.	☐	☐
R/R: Decelerates and shifts gears before the crossing.	☐	☐
R/R: Does not stop, shift, or change lanes while crossing.	☐	☐
Signs: Identifies height clearance signs on route	☐	☐

12. General Vehicle Control and Passenger Safety:	Go	No Go
Understands GVW, weight balance, and effects of overloading	☐	☐
Uses seat belts; follows safe loading/unloading and light protocols	☐	☐
Verifies surroundings; uses mirrors and spotters; controls speed	☐	☐

13. Remarks:

SECTION IV. CERTIFIERS ENDORSEMENT

I, the undersigned, do attest and certify that the trainee named herein has successfully completed all required training and is deemed proficient in the operation of the vehicle(s)/equipment listed in **Section II**. This certification is made to the best of my knowledge and belief, based on training conducted pursuant to the approved qualification training package.

14. Signature	**15. Date**

Attachment 4

SEVEN-STEP INSPECTION PROCESS

Figure A4.1. Seven-Step Inspection Process.

SEVEN-STEP VEHICLE INSPECTION GUIDE

Step 1: Vehicle Overview

Review Last Vehicle Inspection Report. Drivers may have to make a vehicle inspection report in writing each day. The motor carrier must repair any items in the report that affect safety and certify on the report that repairs were made or were unnecessary. Only sign the previous report after verifying that any noted defects have been repaired or certified as not needing repair..

Step 2: Check Engine Compartment

Parking Brakes Are On and/or Wheels Chocked.

You may have to raise the hood, tilt the cab (ssecure any loose items to prevent them from falling and causing damage), or open the engine compartment door.

Check the following:

- Engine oil level.
- Coolant level in radiator; condition of hoses.

Check That the

- Power steering fluid level; hose condition (if so equipped).
- Windshield washer fluid level

Battery fluid level, connections, and tie downs (battery may be located elsewhere)

- Automatic transmission fluid level (may require engine to be running).
- Check belts for tightness and excessive wear (alternator, water pump, air compressor) --learn how much "give" the belts should have when adjusted right and check each one.
- Leaks in the engine compartment (fuel, coolant, oil, power steering fluid, hydraulic fluid, battery fluid).
- Cracked, worn electrical wiring insulation.
- Lower and secure hood, cab, or engine compartment door.

Step 3: Start Engine and Inspect Inside the Cab

Get In and Start Engine

Make sure parking brake is on.

Put gearshift neutral (or "park" if automatic).

Start engine; listen for unusual noises.

If equipped, check the Anti-lock Braking System (ABS) indicator lights. Light on dash should come on and then turn off. If it stays on the ABS is not working properly. For trailers only, if the yellow light on the left rear of the trailer stays on, the ABS is not working properly.

Look at the Gauges

Oil pressure. Should come up to normal within seconds after engine is started. See Figure 2.5

Air pressure. Pressure should build from 50 to 90 psi within 3 minutes. Build air pressure to governor cut-out (usually around 120 – 140 psi. **Know your vehicle's requirements.**

Ammeter and/or voltmeter. Should be in normal range(s).

Coolant temperature. Should begin gradual rise to normal operating range.

Engine oil temperature. Should begin gradual rise to normal operating range.

Warning lights and buzzers. Oil, coolant, charging circuit warning, and antilock brake system lights should go out right away.

Check Condition of Controls. Check all of the following for looseness, sticking, damage, or improper setting:

- Steering wheel.
- Clutch.
- Accelerator ("gas pedal").
- Brake controls.
- Foot brake.
- Trailer brake (if vehicle has one).
- Parking brake.
- Retarder controls (if vehicle has them).
- Transmission controls.
- Interaxle differential lock (if vehicle has one).
- Horn(s).
- Windshield wiper/washer.
- Lights.
- Headlights.
- Dimmer switch.
- Turn signal.
- Four-way flashers.
- Parking, clearance, identification, marker switch(es).

OIL PRESSURE

- Idling 5-20 PSI
- Operating 35-75 PSI
- Low, Dropping, Fluctuating: **STOP IMMEDIATELY!** Without oil the engine can be destroyed rapidly

Figure 2.5

Check Mirrors and Windshield. Inspect mirrors and windshield for cracks, dirt, illegal stickers, or other obstructions to seeing clearly. Clean and adjust as necessary.

Check Emergency Equipment

Check for safety equipment:

Spare electrical fuses (unless vehicles have circuit breakers).

Three red reflective triangles, 6 fuses or 3 liquid burning flares.

Properly charged and rated fire extinguisher.

Check for optional items such as:

- Chains (where winter conditions require).
- Tire changing equipment.
- List of emergency phone numbers
- Accident reporting kit (packet).

Check Safety Belt. Check that the safety belt is securely mounted, adjusts; latches properly and is not ripped or frayed.

Step 4: Turn Off Engine and Check Lights

Make sure the parking brake is set, turn off the engine, and take the key with you. Turn on headlights (low beams) and four-way emergency flashers and get out of the vehicle.

Step 5: Do Walk-around Inspection

- Go to front of vehicle and check that low beams are on, and both of the four-way flashers are working.
- Push dimmer switch and check that high beams work.
- Turn off headlights and four-way emergency flashers.
- Turn on parking, clearance, side-marker, and identification lights.
- Turn on right turn signal and start walk-around inspection.

General

- Walk around and inspect.
- Clean all lights, reflectors, and glass as you go along.

Left Front Side

- Driver's door glass should be clean.
- Door latches or locks should work properly.
- Left front wheel.
- Condition of wheel and rim--missing, bent, broken studs, clamps, lugs, or any signs of misalignment.
- Condition of tires--properly inflated, valve stem and cap OK, no serious cuts, bulges, or tread wear.
- Use wrench to test rust-streaked lug nuts, indicating looseness.
- Hub oil level OK, no leaks.
- Left front suspension.
- Condition of spring, spring hangers, shackles, U-bolts.
- Shock absorber condition.
- Left front brake.
- Condition of brake drum or disc.
- Condition of hoses.

Front

- Condition of front axle and steering system.
- No loose, worn, bent, damaged or missing parts.
- Must grab steering mechanism to test for looseness.
- Condition of windshield.
- Check for damage and clean if dirty.
- Check windshield wiper arms for proper spring tension.
- Check wiper blades for damage, "stiff" rubber, and securement.
- Lights and reflectors.
- Parking, clearance, and identification lights clean, operating, and proper color (amber at front).
- Reflectors clean and proper color (amber at front).
- Right front turn signal light clean, operating, and proper color (amber or white on signals facing forward).

Right Side

- Right front: check all items as done on left front.
- Primary and secondary safety cab locks engaged (if cab-over-engine design).
- Right fuel tank(s).
- Securely mounted, not damaged, or leaking.
- Fuel crossover line secure.
- Tank(s) contain enough fuel.
- Cap(s) on and secure.
- Condition of visible parts.
- Rear of engine--not leaking.
- Transmission--not leaking.
- Exhaust system--secure, not leaking, not touching wires, fuel, or air-lines.
- Frame and cross members--no bends or cracks.
- Air-lines and electrical wiring--secured against snagging, rubbing, wearing.
- Spare tire carrier or rack not damaged (if so equipped).
- Spare tire and/or wheel securely mounted.
- Spare tire and wheel adequate (proper size, properly inflated).
- Cargo securement (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Header board adequate, secure
- Side boards, stakes strong enough, free of damage, properly set in place
- Canvas or tarp properly secured to prevent tearing, billowing, or blocking of mirrors.
- If oversize, all required signs (flags, lamps, and reflectors) safely and properly mounted and all required permits in driver's possession.
- Curbside cargo compartment doors in good condition, securely closed, latched/locked and required security seals in place.

Right Rear

- Condition of wheels and rims--no missing, bent, or broken spacers, studs, clamps, or lugs.
- Condition of tires--properly inflated, valve stems and caps OK, no serious cuts, bulges, tread wear, tires not rubbing each other, and nothing stuck between them.
- Tires same type, e.g., not mixed radial and bias types.
- Tires evenly matched (same sizes).
- Wheel bearing/seals not leaking.
- Suspension.
- Condition of spring(s), spring hangers, shackles, and U-bolts.
- Axle secure.
- Powered axle(s) not leaking lube (gear oil).
- Condition of torque rod arms, bushings.
- Condition of shock absorber(s).
- If retractable axle equipped, check condition of lift mechanism. If air powered, check for leaks.
- Condition of air ride components.
- Brakes.
- Brake adjustment.
- Condition of brake drum(s) or discs.
- Condition of hoses--look for any wear due to rubbing.
- Lights and reflectors.
- Side-marker lights clean, operating, and proper color (red at rear, others amber).
- Side-marker reflectors clean and proper color (red at rear, others amber).

Rear

- Lights and reflectors.
- Rear clearance and identification lights clean, operating, and proper color (red at rear).
- Reflectors clean and proper color (red at rear).
- Taillights clean, operating, and proper color (red at rear).
- Right rear turn signal operating, and proper color (red, yellow, or amber at rear).
- License plate(s) present, clean, and secured.
- Splash guards present, not damaged, properly fastened, not dragging on ground, or rubbing tires.
- Cargo secure (trucks).
- Cargo properly blocked, braced, tied, chained, etc.
- Tailboards up and properly secured.
- End gates free of damage, properly secured in stake sockets.
- Canvas or tarp (if required) properly secured to prevent tearing, billowing, or blocking of either the rearview mirrors or rear lights.
- If over-length, or over-width, make sure all signs and/or additional lights/flags are safely and properly mounted and all required permits are in driver's possession.
- Rear doors closed, latched/locked.

Left Side

- Check all items as done on right side, plus:
- Battery(ies)
- Battery box(es) securely mounted.
- Box has secure cover.
- Battery(ies) secured against movement.
- Battery(ies) not broken or leaking.
- Fluid in battery(ies) at proper level (except maintenance-free type).
- Cell caps present and securely tightened (except maintenance-free type).
- Vents in cell caps free of foreign material (except maintenance-free type).

Step 6: Check Signal Lights

- Get In and Turn Off Lights
- Turn off all lights.
- Turn on stop lights (apply trailer hand brake or have a helper put on the brake pedal).
- Turn on left turn signal lights.
- Get Out and Check Lights
- Left front turn signal light clean, operating and proper color (amber or white on signals facing the front).
- Left rear turn signal light and both stop lights clean, operating, and proper color (red, yellow, or amber).

Get In Vehicle

- Turn off lights not needed for driving.
- Check for all required papers, trip manifests, permits, etc.
- Secure all loose articles in cab (they might interfere with operation of the controls or hit you in a crash).
- Start the engine.

Step 7: Start the Engine and Check

Test for Hydraulic Leaks. If the vehicle has hydraulic brakes, pump the brake pedal three times. Then apply firm pressure to the pedal and hold for five seconds. The pedal should not move.

If it does, there may be a leak or other problem. Get it fixed before driving. If the vehicle has air brakes, do the checks described in Sections 5 and 6 of this manual.

Brake System

Test Parking Brake(s)

- Fasten safety belt
- Set parking brake (power unit only).
- Release trailer parking brake (if applicable).
- Place vehicle into a low gear.
- Gently pull forward against parking brake to make sure the parking brake holds.
- Repeat the same steps for the trailer with trailer parking brake set and power unit parking brakes released (if applicable).
- If it doesn't hold vehicle, it is faulty; get it fixed.

Test Service Brake Stopping Action

- Go about five miles per hour.
- Push brake pedal firmly
- "Pulling" to one side or the other can mean brake trouble.
- Any unusual brake pedal "feel" or delayed stopping action can mean trouble.
- If any defects are found during the inspection, they must be repaired before operating the vehicle. Federal and state laws prohibit driving an unsafe vehicle.